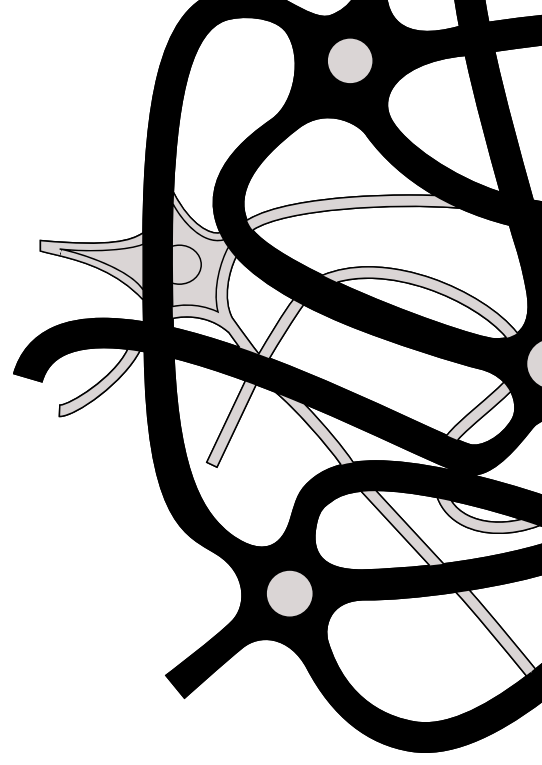
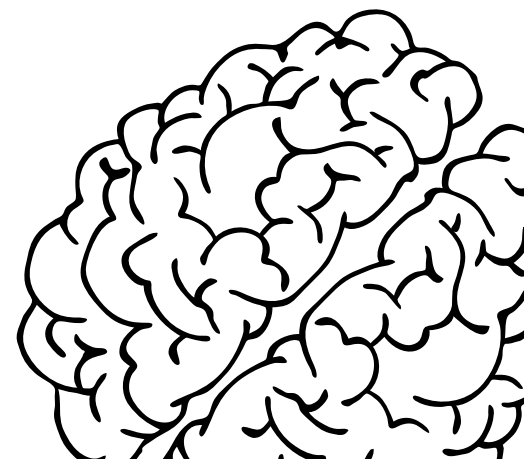
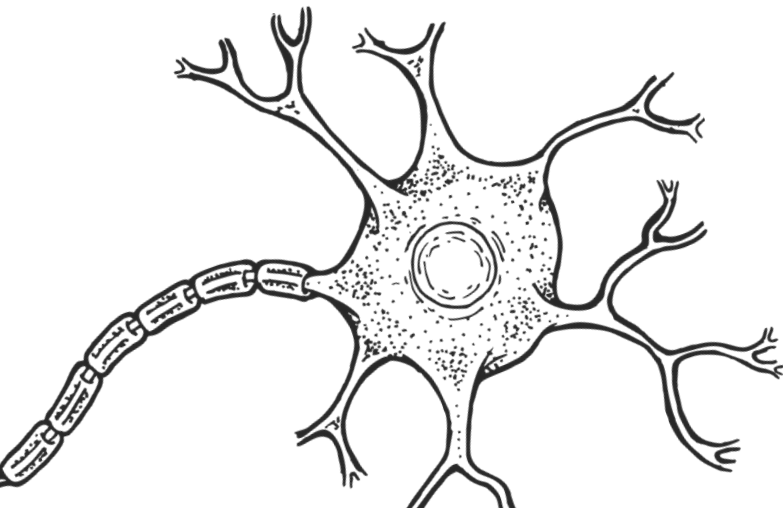
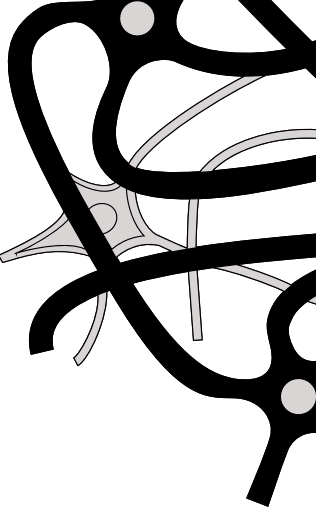
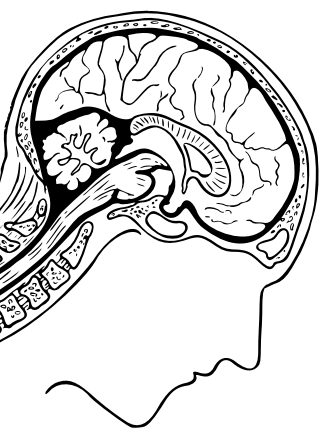




Session 3: EEG Hardware & Signal Acquisition

Presented by Nada

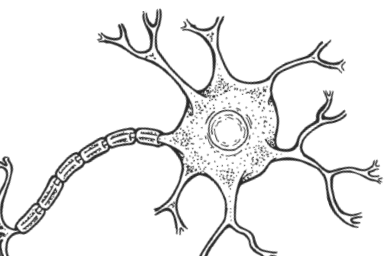
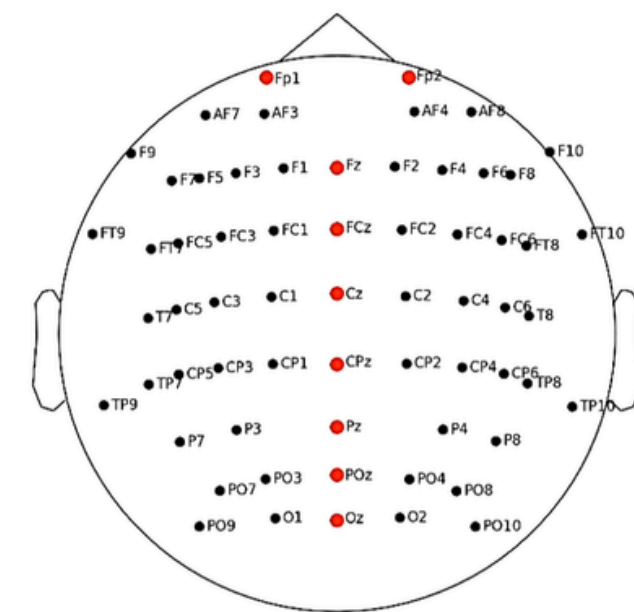


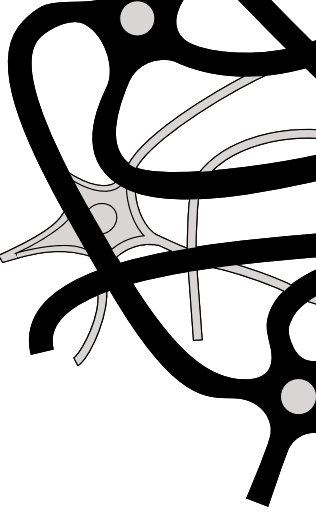
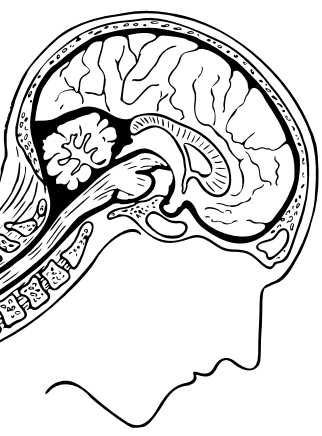


Goals

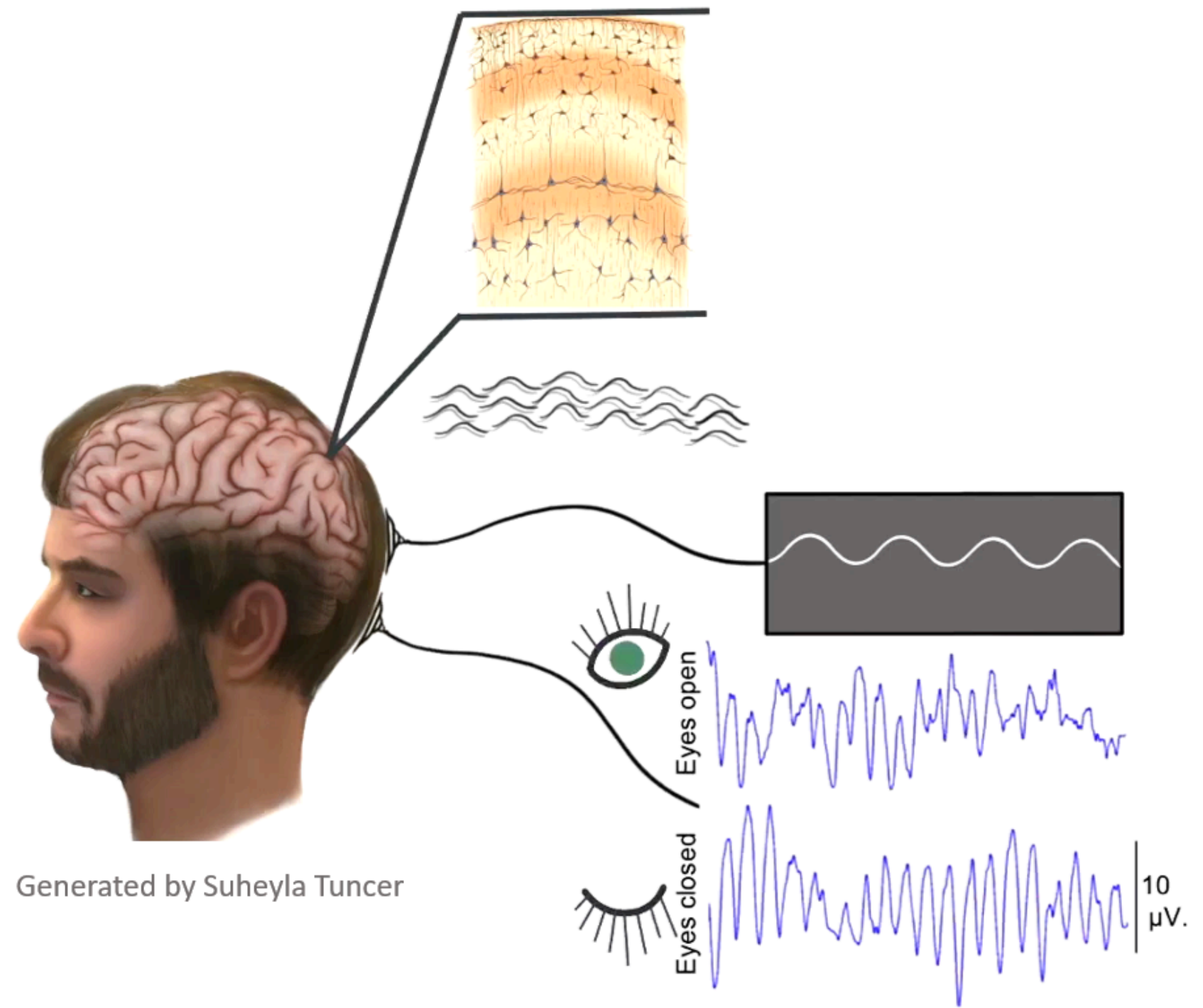
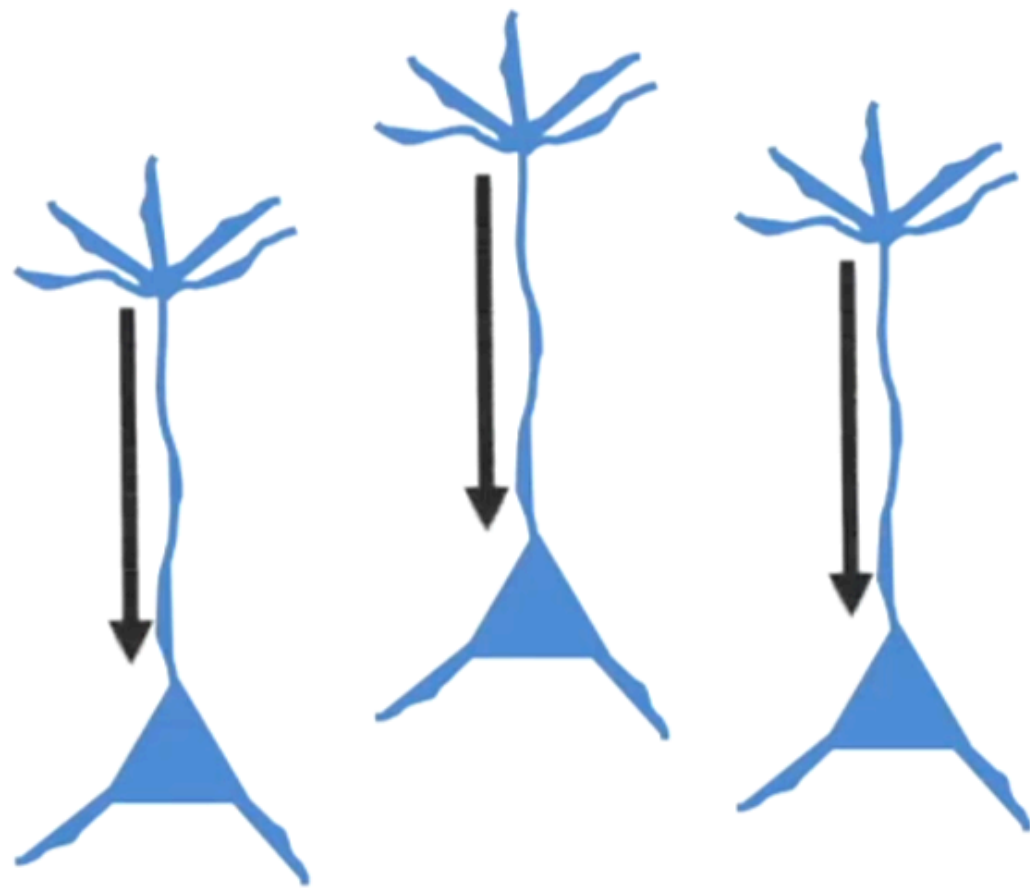
By the end of this session, you should be able to:

- Describe the physical components of an EEG system
- Explain the electrode-skin interface and what impedance means
- Read and use the 10–20 International System of electrode placement

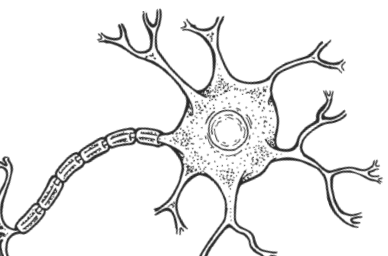


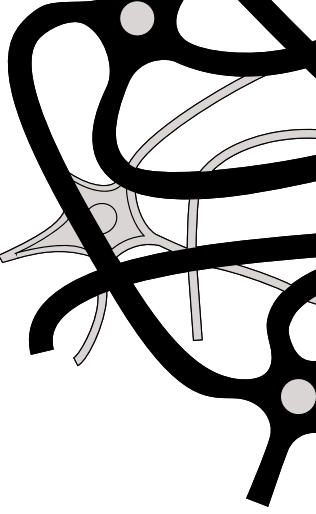
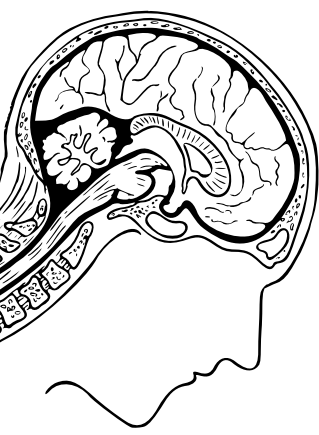


Recap: Origin of EEG Signal



Generated by Suheyly Tuncer



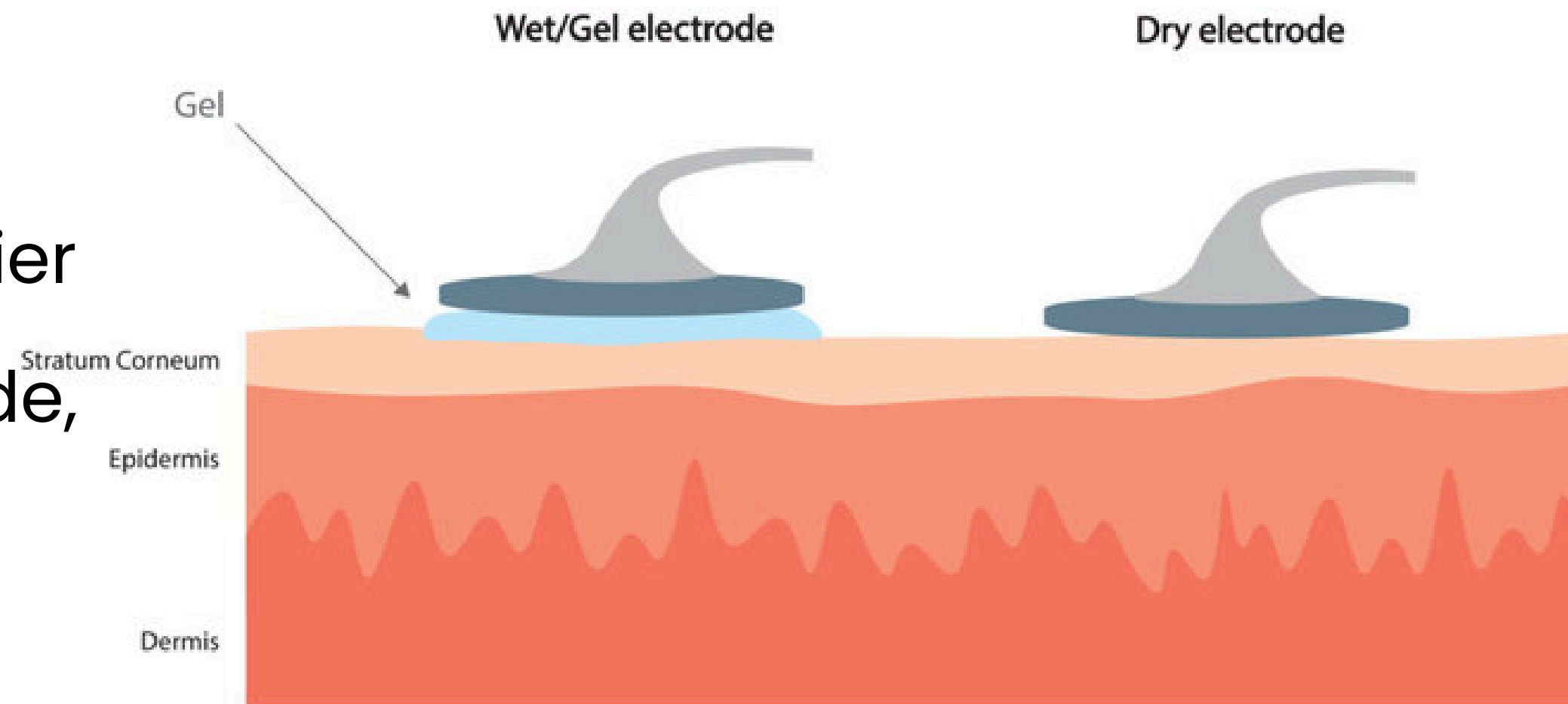


Electrodes

The interface between brain and machine

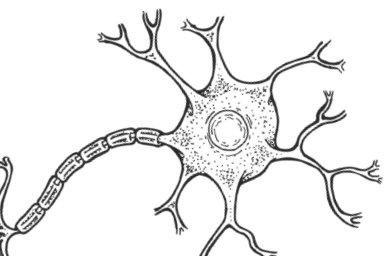
By Type

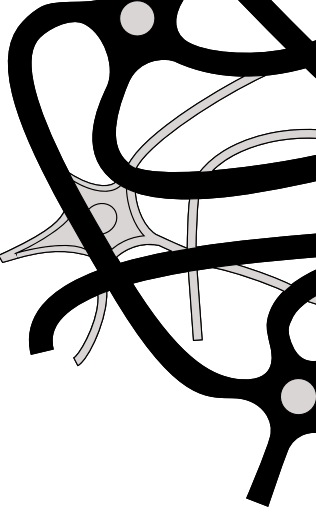
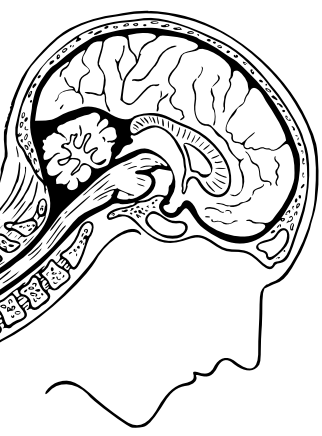
- **Passive:** simple metal, amplifier is remote
- **Active:** amp built into electrode, less cable noise
- **Wet:** conductive gel, lower impedance
- **Dry:** no prep needed, higher impedance



The gel's job?

1. Bridges the electrode to skin gap
2. Lowers impedance → better signal
3. Keeps contact stable during movement





Electrodes

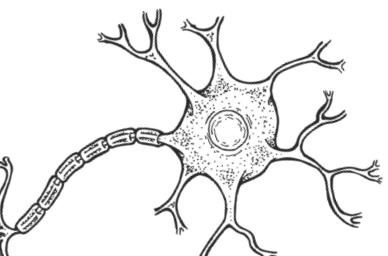
The interface between brain and machine

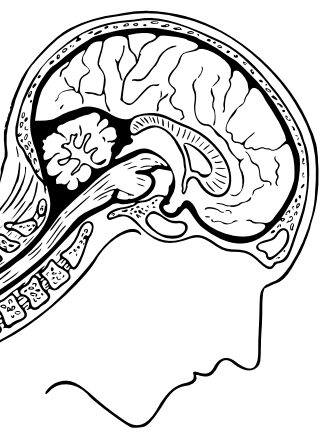
By Material

- **Ag/AgCl:** gold standard, low noise, stable DC offset
- Gold/tin: older designs, less common now

Cap vs individual

- Cap → fast setup, consistent layout, good for research
- Individual → flexible placement, slower to prep





The 10-20 International System

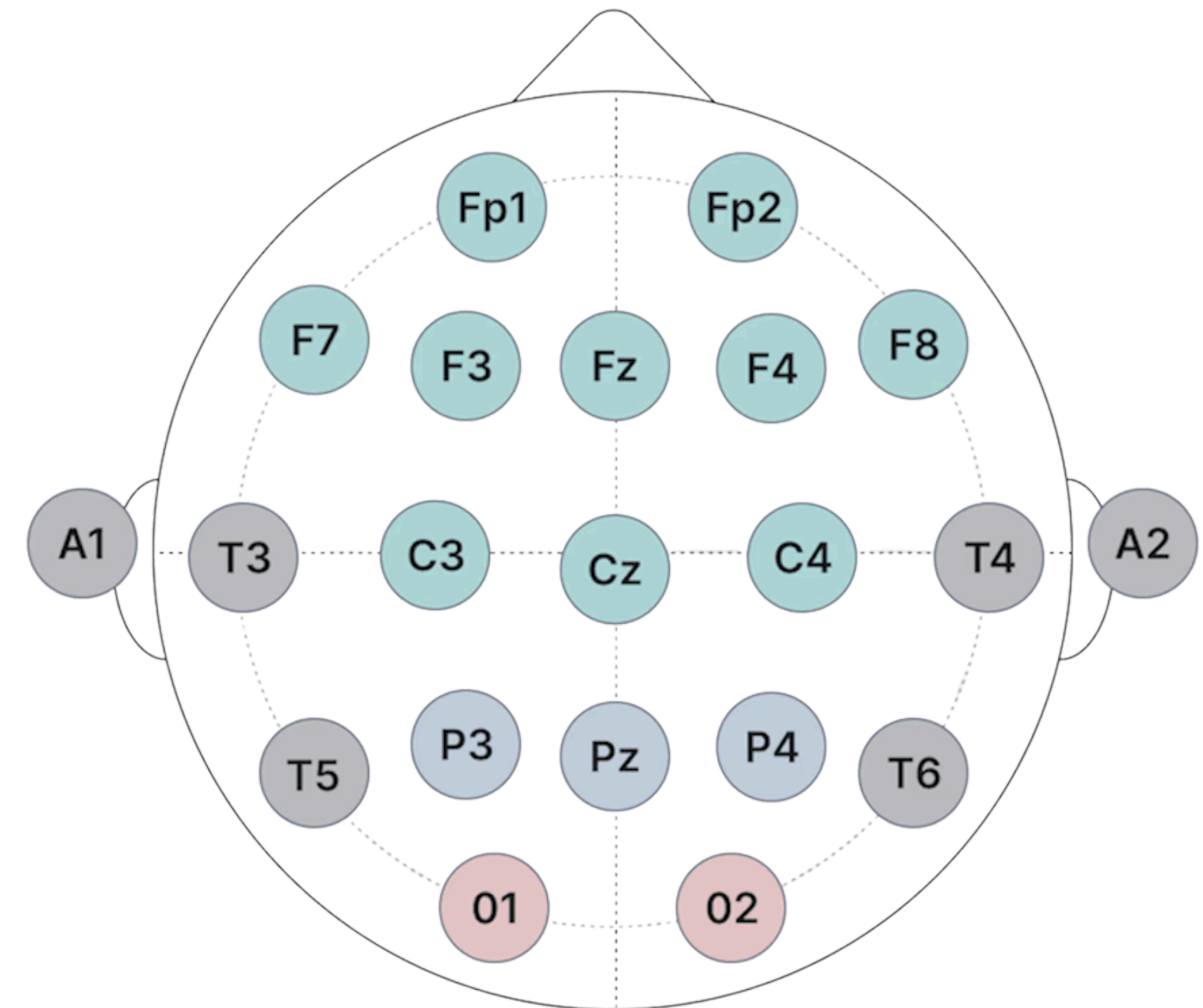
A standardised map so labs worldwide speak the same language

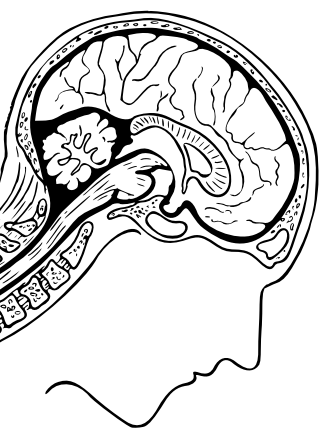


Herbert H. Jasper

Why 10-20?

- Developed by Jasper & colleagues, 1958
- Electrodes spaced 10% or 20% of skull circumference apart
- Proportions works for any head size
- Reproducible through labs and countries





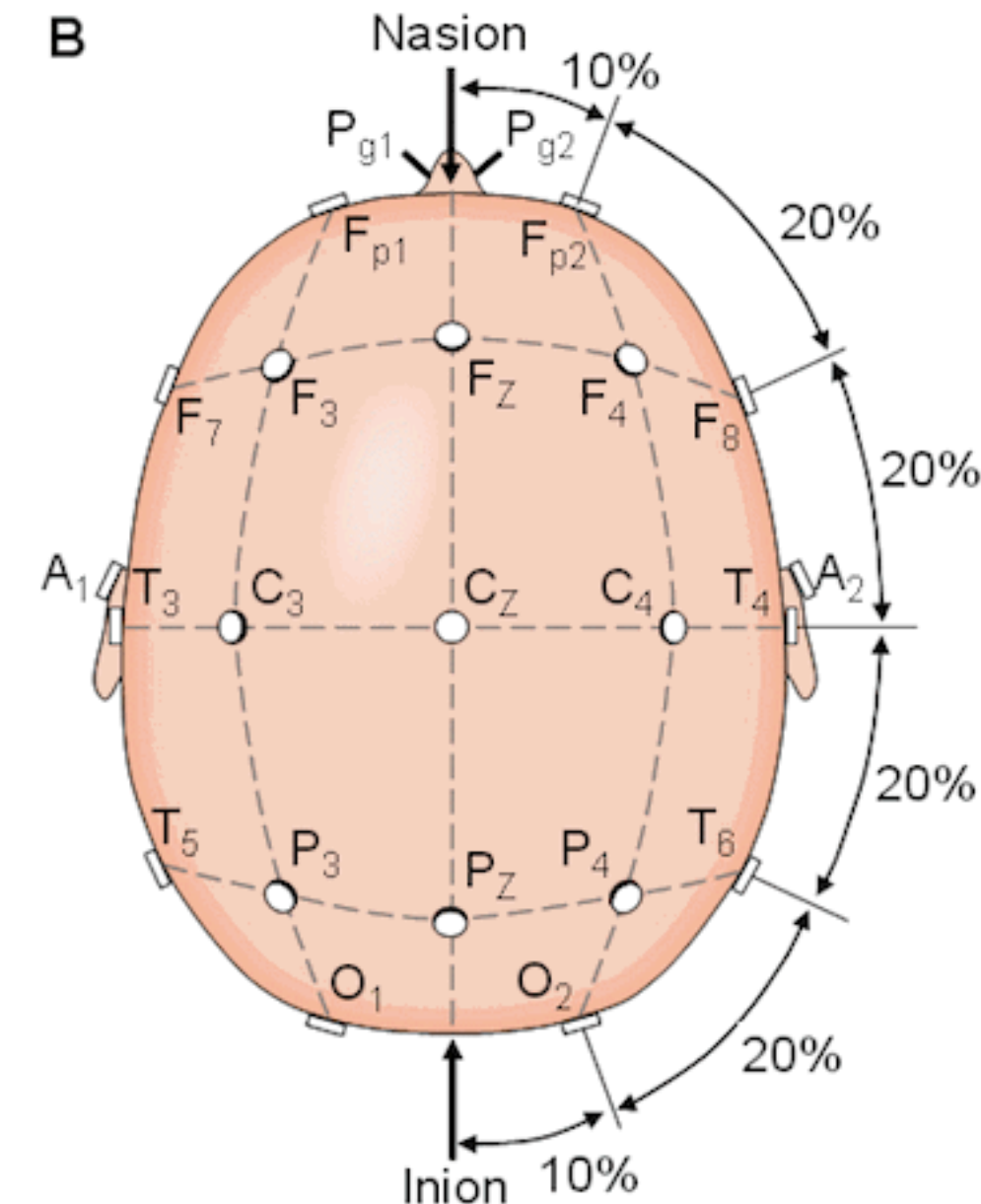
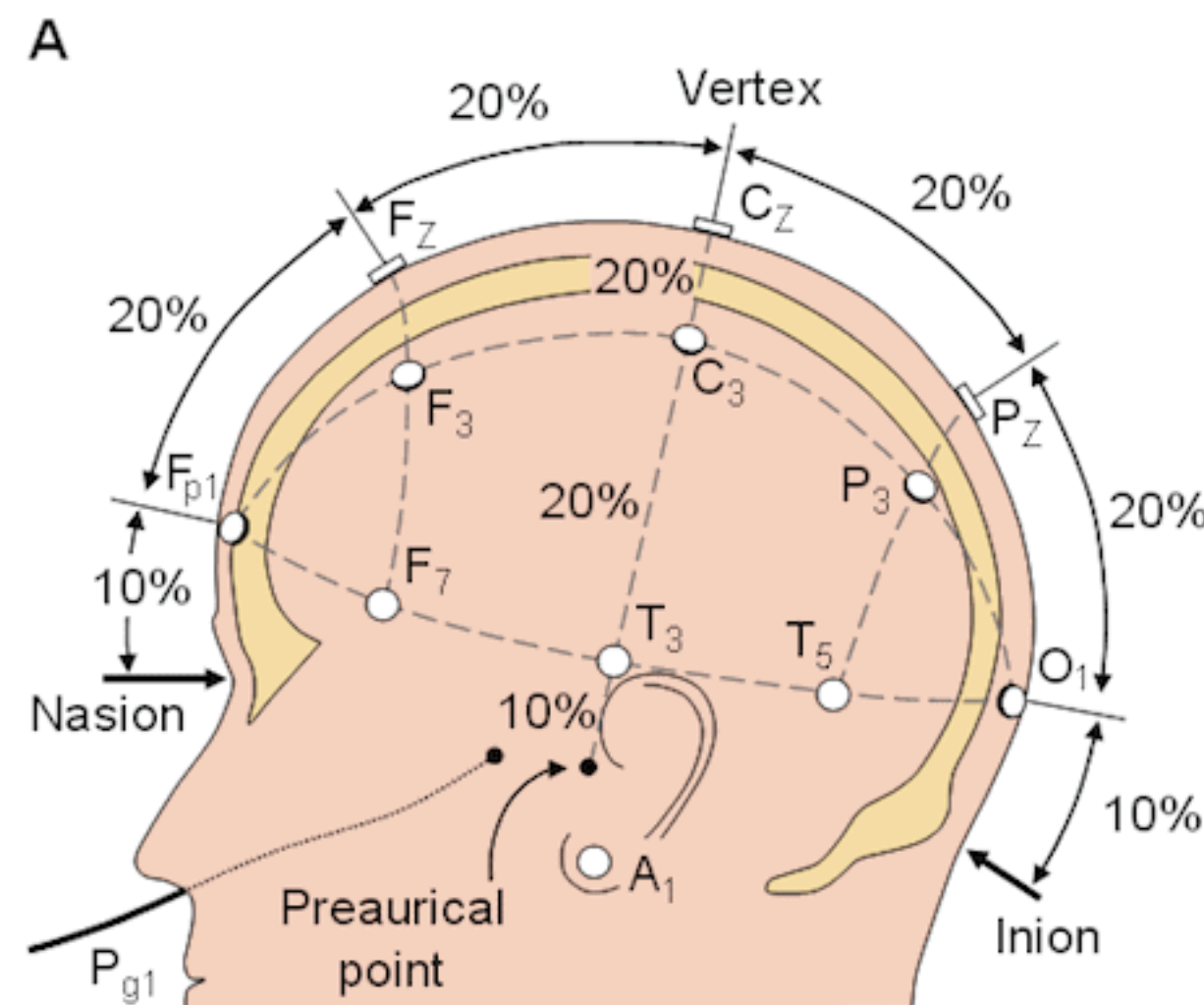
The 10-20 International System

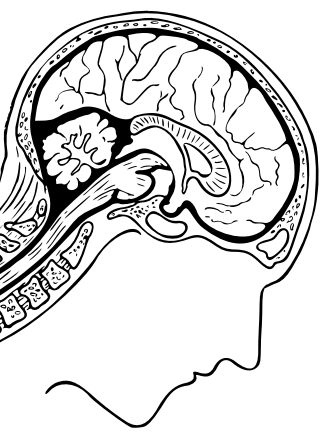
Naming logic

- Letter = brain region (Fp, F, C, P, O, T)
- Number = odd = left hemisphere, even = right
- z = midline (Fz, Cz, Pz, Oz)

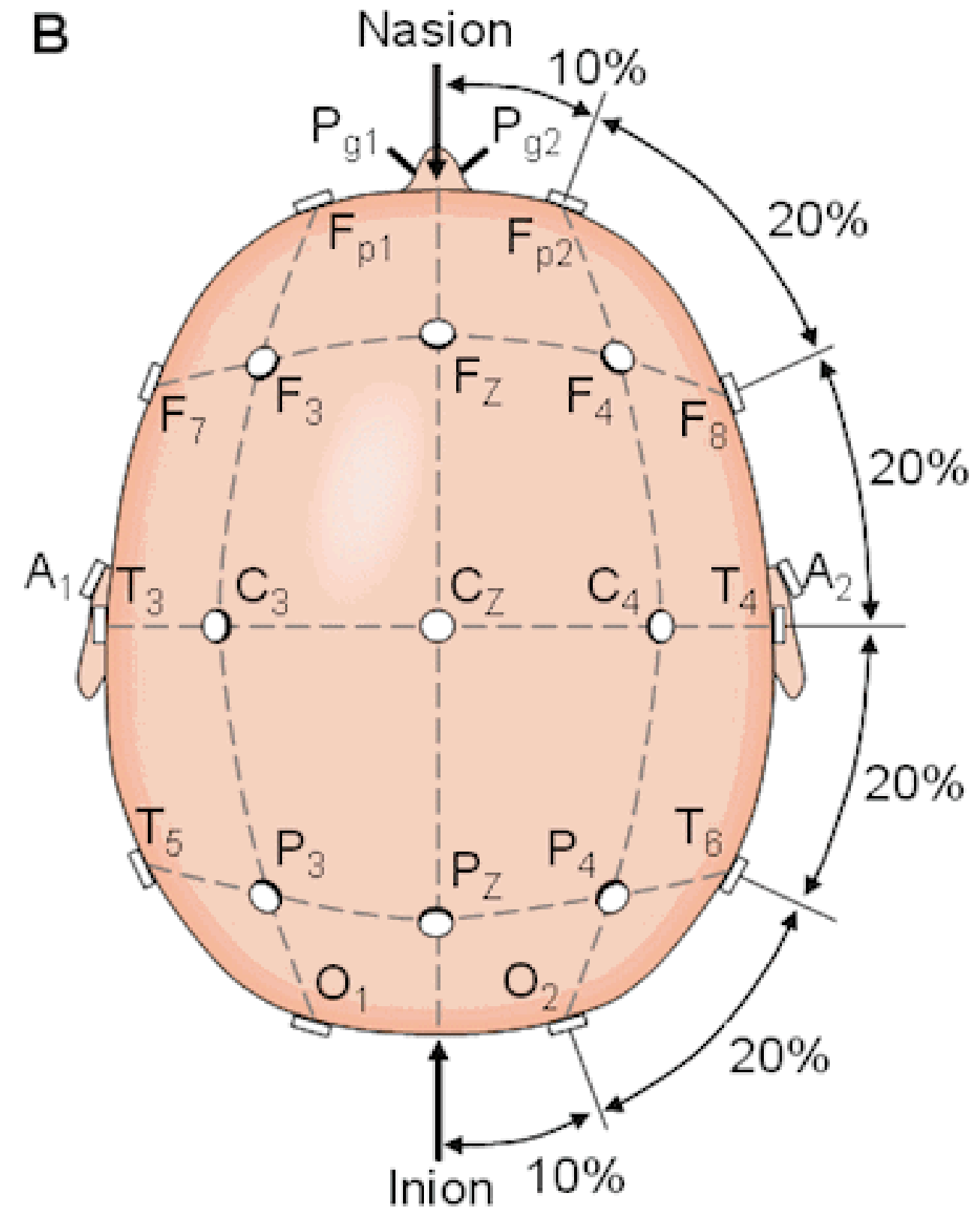
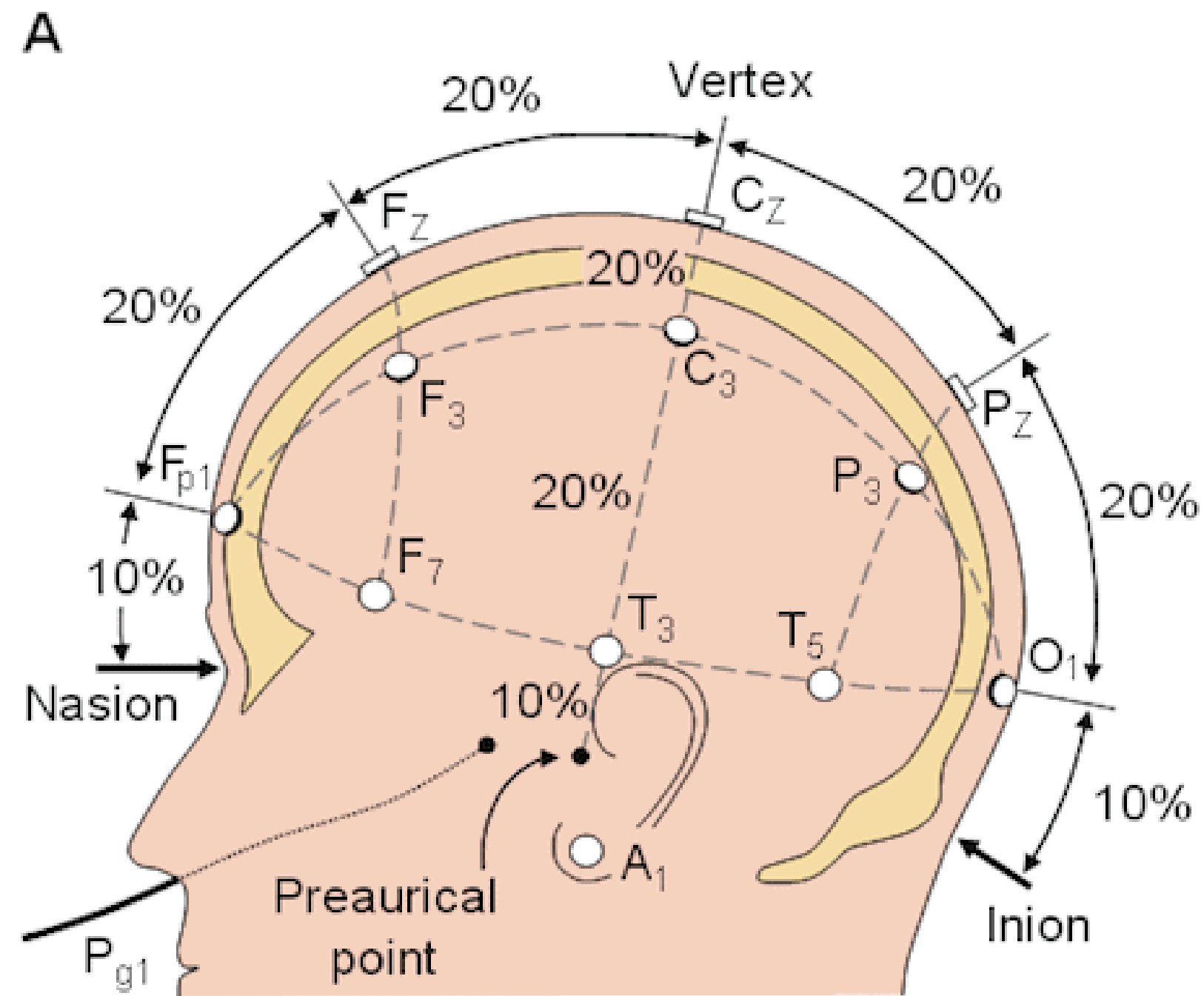
Regions

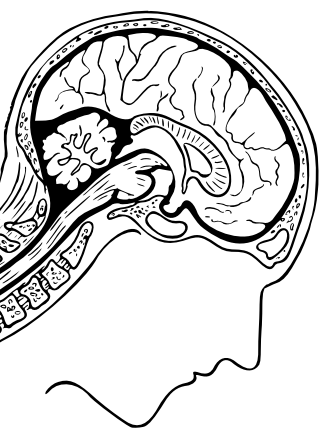
- **Fp** = frontal pole (eye, forehead)
- **F** = frontal (executive function)
- **C** = central (motor, somatosensory)
- **P** = parietal (attention, space)
- **O** = occipital (vision)
- **T** = temporal (language, memory)



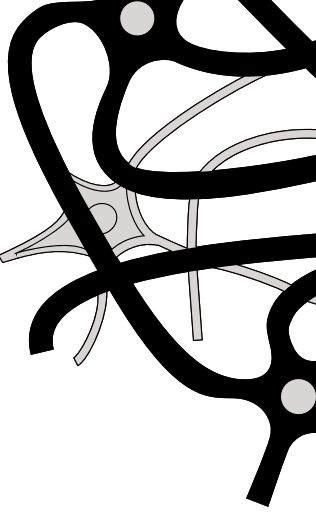


Why electrode location matters?



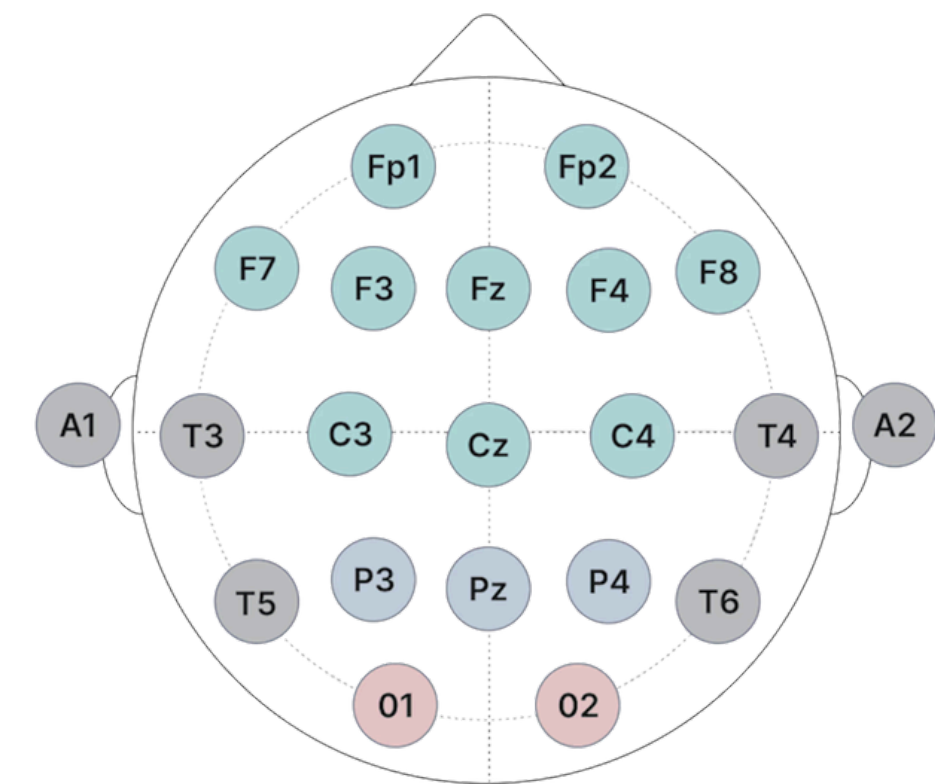


The scalp is a window where you look determines what you see



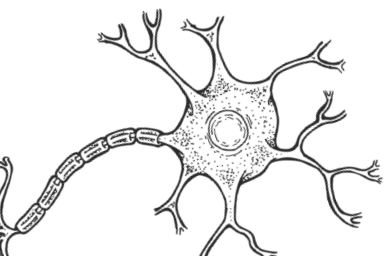
Practical Rule

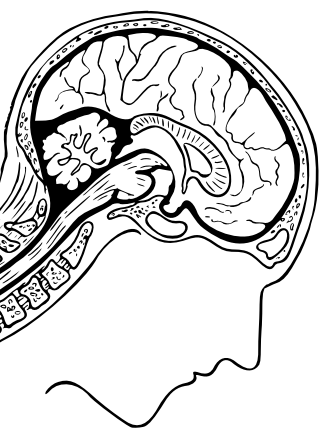
- If your task involves vision: report O channels
- If language: left temporal (T7, F7)
- If motor: C3/C4 contralateral to hand
- If attention/P300: Pz is your primary channel



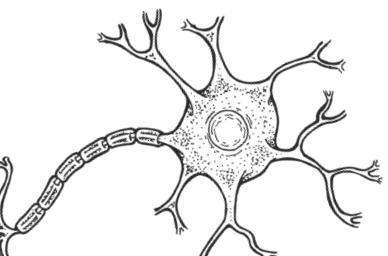
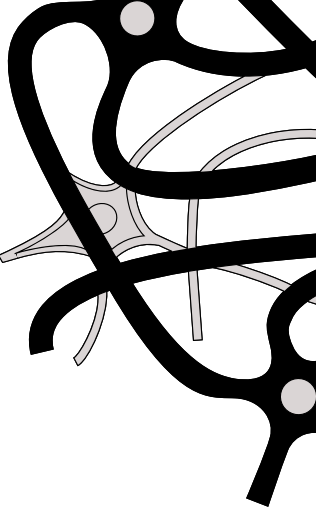
High-density systems

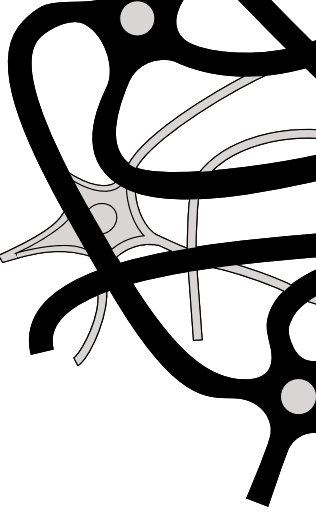
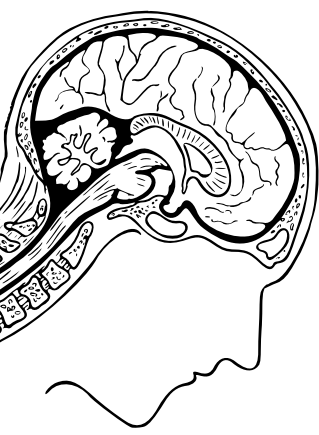
- 32-ch for basic cognitive research
- 64-ch for standard research grade
- 128-ch for source localisation possible
- 256-ch+ for clinical HD-EEG





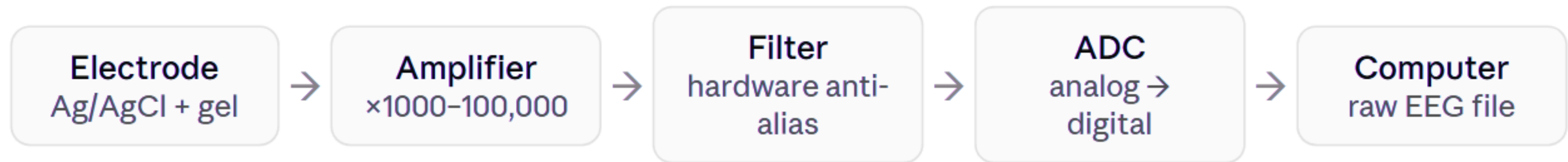
10-20 map - tap any electrode!





The signal chain

What happens before you see data

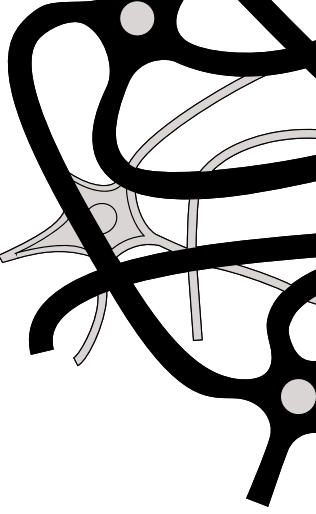
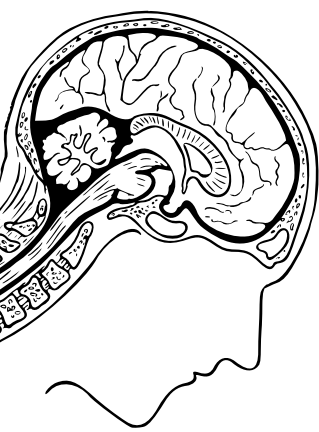


Amplifier:

1. Gain: how much the signal is boosted (EEG is tiny: $\sim 10-100 \mu\text{V}$)
1. CMRR: common-mode rejection ratio; how well it cancels noise shared across all channels (line noise, interference)

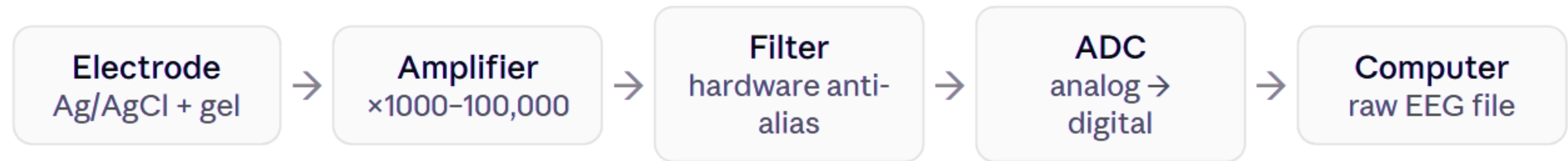
Reference vs ground

1. Reference: every EEG channel is a voltage difference between the active electrode and a reference point
2. Ground: separate electrode that reduces electrical interference



The signal chain

What happens before you see data

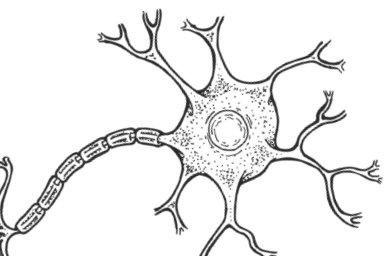


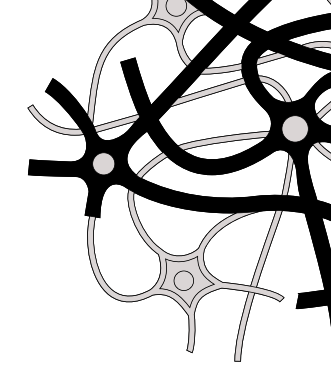
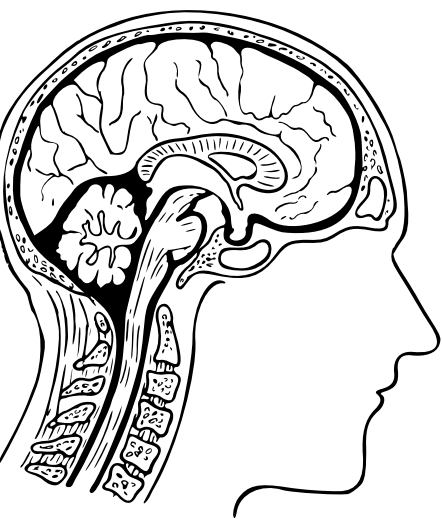
ADC sampling rates

250–500 Hz
budget/portable systems

1000–1024 Hz
standard research

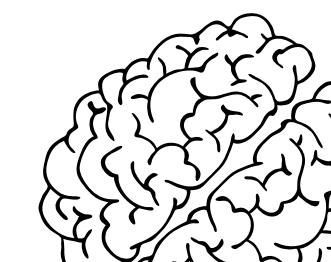
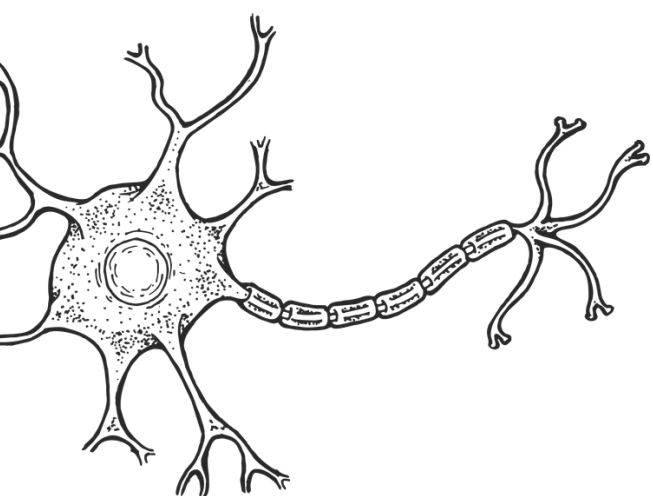
2048 Hz+
high-density / clinical

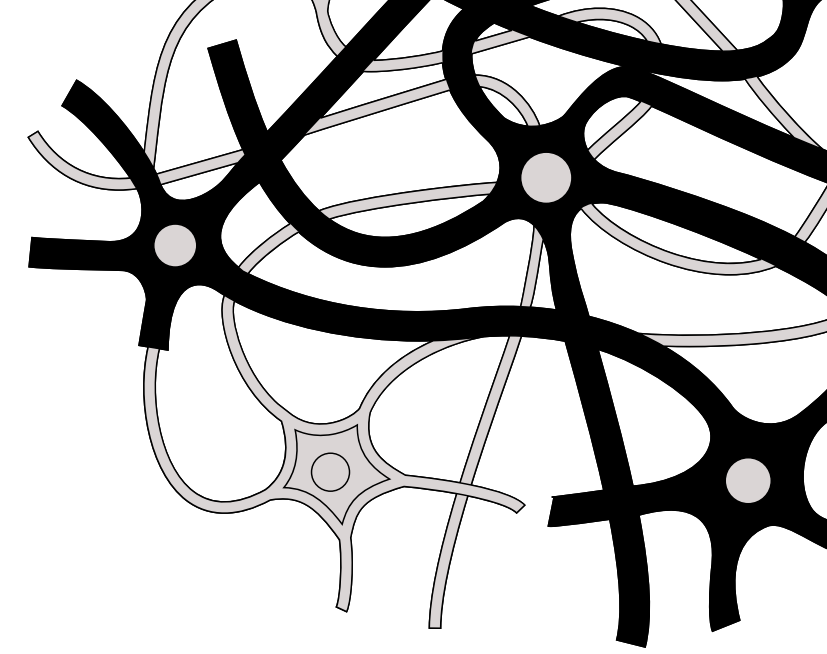
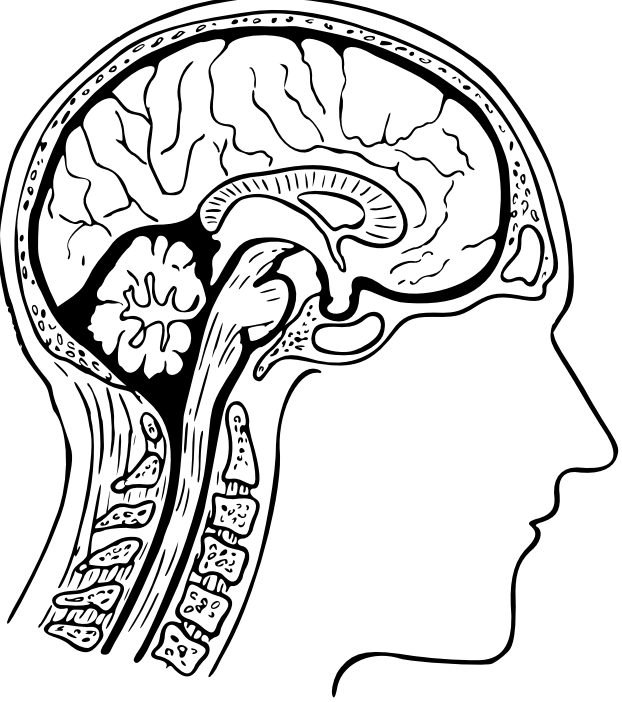




Quick check-in

- » What's the difference between the reference electrode and the ground?
- » If I want to study the P300 component, which channel is my primary target?
- » Why do we use Ag/AgCl and not just any metal?





Questions?

